Local Income Taxes and Residential Sorting:

Evidence from LEHD Origin-Destination Data

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Abstract

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1 Introduction

2 Institutional Background

2.1 Ohio Municipal Income Tax

2.2 Pennsylvania Earned Income Tax

3 Data

3.1 LODES Origin-Destination Data

3.2 Municipal Tax Rate Panels

3.3 Municipal Financial Reports

3.4 Outcome Variable: share_live_in

Table 1: Summary Statistics

Variable	Mean	Median	SD	Min	Max
Panel A: Pennsylvania (2002–2019)					
share_live_in	0.149	0.127	0.097	0	0.700
Resident EIT rate (%)	0.542	0.500	0.199	0	4.331
Total jobs	3,624	1,224			
Resident workers	674	154			
Panel B: Ohio (2002–2019)					
share_live_in	0.119	0.099	0.085		
Municipal tax rate (%)	1.222	1.500	0.821	0	3.000
Panel dimensions					
PA municipality × years	24,528 (1,383 municipalities)				
OH municipality × years	7,718 (430 municipalities)				

Notes: Municipality-year panel, 2002–2019. Sample restricted to municipalities with > 50 total jobs. share_live_in = workers who live and work in municipality / total workers in municipality. OH municipalities are Census places; PA municipalities are county subdivisions.

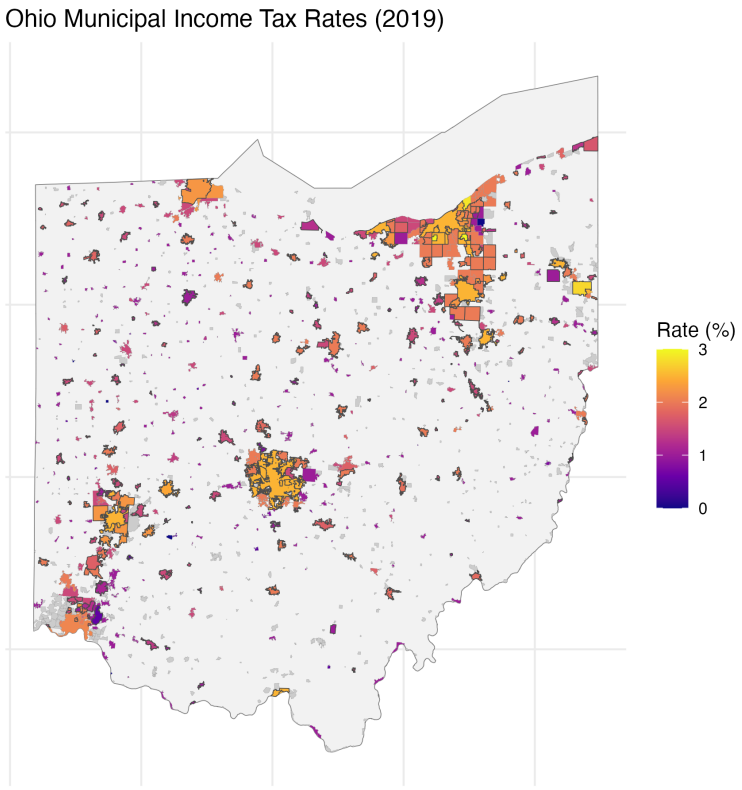


Figure 1: Ohio Municipal Income Tax Rates (2019). Bold outline = municipalities that raised rates during the sample period. Grey fill = municipalities with no municipal income tax.

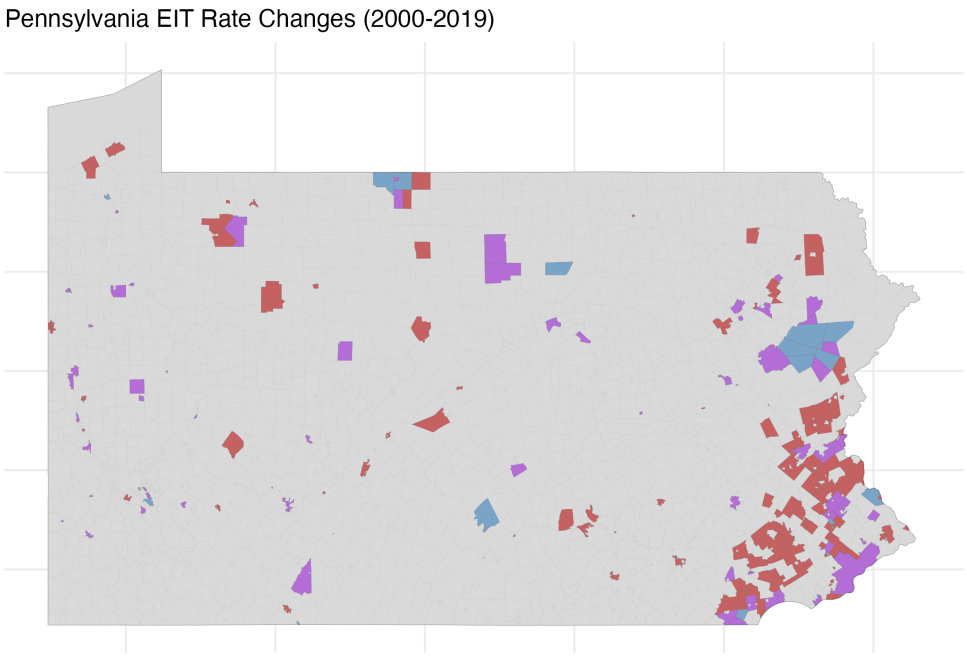


Figure 2: Pennsylvania EIT Rate Changes (2000–2019). Red = increased; blue = decreased; purple = both; grey = no change.

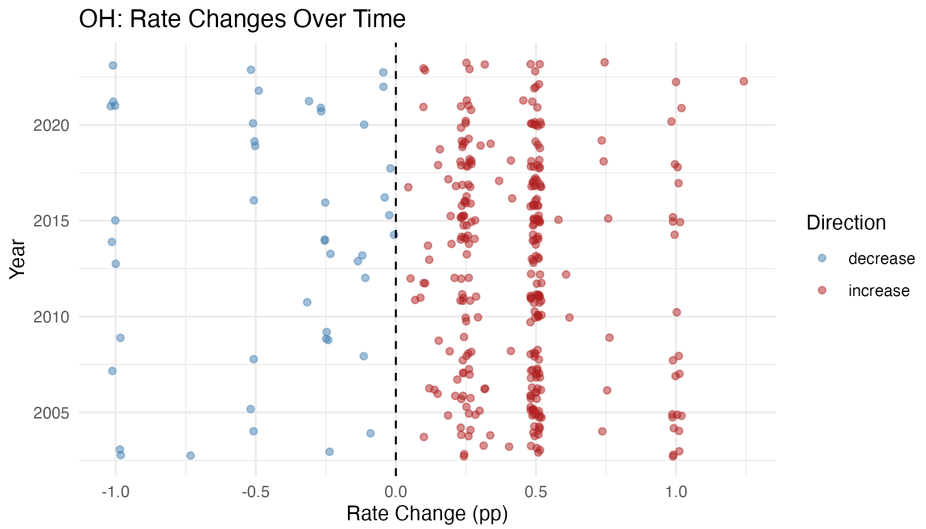


Figure 3: Ohio: Distribution of Municipal Income Tax Rate Changes Over Time

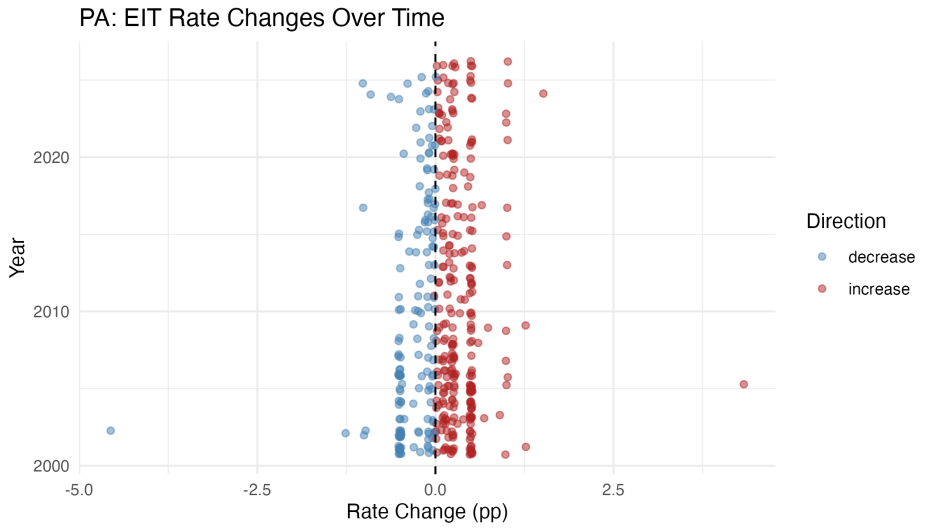


Figure 4: Pennsylvania: Distribution of EIT Rate Changes Over Time

4 Empirical Strategy

We organize our analysis around four sequential questions, each building on the previous. All specifications use the Sun and Abraham (2021) interaction-weighted estimator to handle staggered treatment timing, with municipality and year fixed effects and standard errors clustered at the municipality level:

$$Y_{it} = \alpha_i + \gamma_t + \sum_{\ell \neq -1} \mu_\ell \cdot D_{it}^{\ell} + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{it} \tag{1}$$

where Y_{it} is share_live_in, D_{it}^{ℓ} indicates municipality i being ℓ years from its first treatment, and $\mathbf{X}_{it}$ is a vector of fiscal controls (available for PA). Pre-treatment coefficients ($\ell < -1$) test parallel trends.

Q1: Does adopting a local income tax cause sorting? Ohio municipalities can adopt or repeal a local income tax; unincorporated townships have no tax. We compare the 5 municipalities that adopted a tax during the panel to the 104 no-tax townships. This is the cleanest Tiebout test: a binary switch from no tax to having one.

Q1b: Does the rate at adoption matter? Among adopters, we test whether higher initial rates produce more sorting by interacting the post-adoption indicator with the rate set at adoption.

Q2: Does raising an existing rate cause additional sorting? Conditional on already having a tax, does increasing the rate push additional residents out? For Ohio, we compare the 252 municipalities that raised their rate to two control groups: (a) 403 stable-rate taxed cities, and (b) 104 no-tax townships. For Pennsylvania, where all municipalities levy an EIT, we compare the 196 rate increasers to the 2,209 municipalities that never changed.

Q3: Is the response symmetric? If rate increases push residents out, do rate decreases pull them back? We estimate Equation 1 with decreasers as the treated group and never-changed as controls. We also test asymmetry jointly:

$$Y_{it} = \alpha_i + \gamma_t + \delta^+ \cdot \text{Post}_{it}^+ + \delta^- \cdot \text{Post}_{it}^- + \mathbf{X}'_{it}\boldsymbol{\phi} + \varepsilon_{it} \tag{2}$$

and with the split cumulative dose specification:

$$Y_{it} = \alpha_i + \gamma_t + \beta^+ \cdot R_{it}^+ + \beta^- \cdot R_{it}^- + \mathbf{X}'_{it}\boldsymbol{\phi} + \varepsilon_{it} \tag{3}$$

where R_{it}^+ and R_{it}^- are cumulative rate increases and decreases from baseline. If residential location decisions are sticky, $|\beta^+| > |\beta^-|$.

5 Results

5.1 Q1: Tax Adoption (Ohio)

Five Ohio municipalities adopted a local income tax during the panel period, compared against 104 no-tax townships. Despite the small treatment group, the adoption effect is the largest and most precisely estimated result in the paper.

Table 2: Q1: Effect of Tax Adoption on share_live_in (OH)

	(1)	(2)
	Sun-Abraham	Binary Post
ATT / post	-0.0174^{***}	-0.0092^{***}
	(0.0026)	(0.0023)
Treated	5 adopters	5 adopters
Control	104 no-tax	104 no-tax
N	1,877	1,877

Notes: Column (1): Sun-Abraham ATT. Column (2): binary post indicator (TWFE, not staggered-corrected). Adopting a local income tax reduces the locally-resident worker share by 1.7 percentage points. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

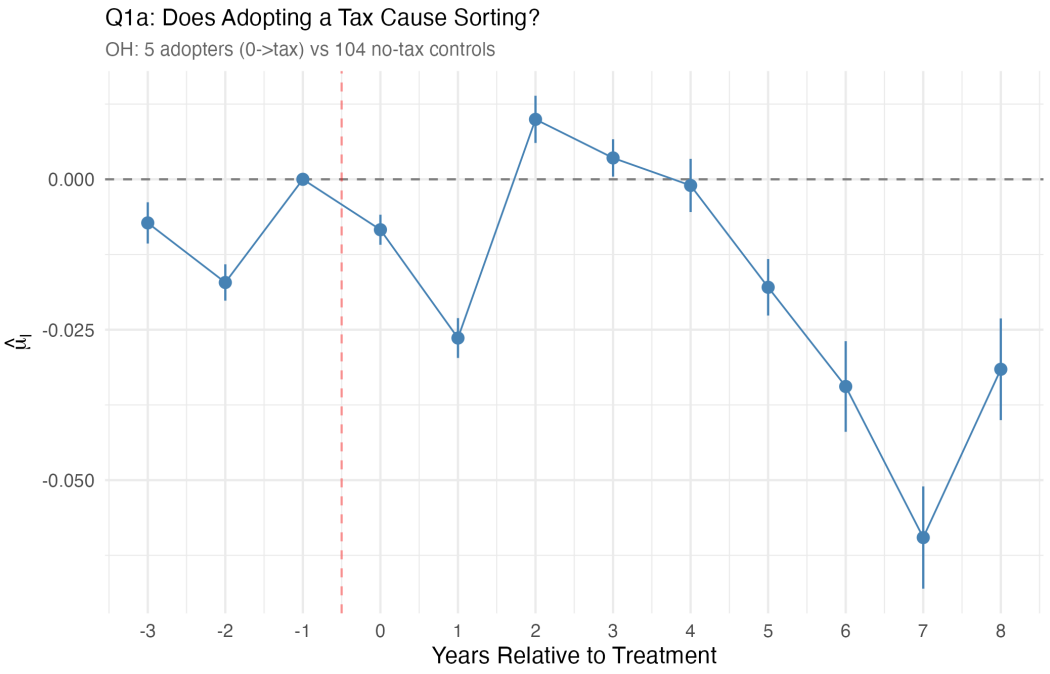


Figure 5: Q1: OH Sun-Abraham Event Study — Tax Adoption (0 → positive) vs No-Tax Controls. 5 adopters, 104 no-tax controls.

5.2 Q2: Rate Increases (Intensive Margin)

Table 3: Q2: Effect of Rate Increases on share_live_in

	(1) OH: Inc vs Stable-Taxed	(2) OH: Inc vs No-Tax	(3) PA: Inc vs Never-Changed
Sun-Abraham ATT	−0.0001	−0.0074**	−0.0053·
	(0.0021)	(0.0025)	(0.0029)
Treated	252 cities	252 cities	196 cities
Control	403 stable	104 no-tax	2,209 NC
N	5,787	5,009	24,257

Notes: Sun-Abraham ATT. Column (1): OH rate increasers vs stable-rate taxed cities (null — poor control group). Column (2): same treated group vs no-tax townships (−0.007**). Column (3): PA rate increasers vs never-changed (−0.005·). $^{\cdot}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

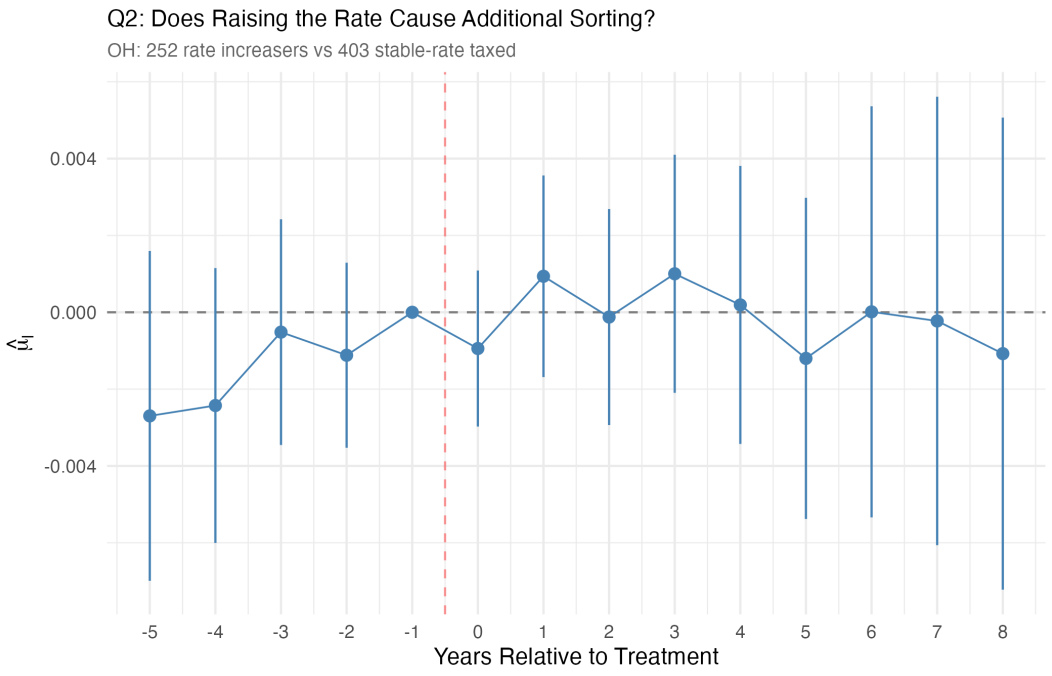


Figure 6: Q2 col (1): OH Rate Increases vs Stable-Rate Taxed. 252 treated, 403 controls. Null result — stable-rate taxed cities are a poor control group.

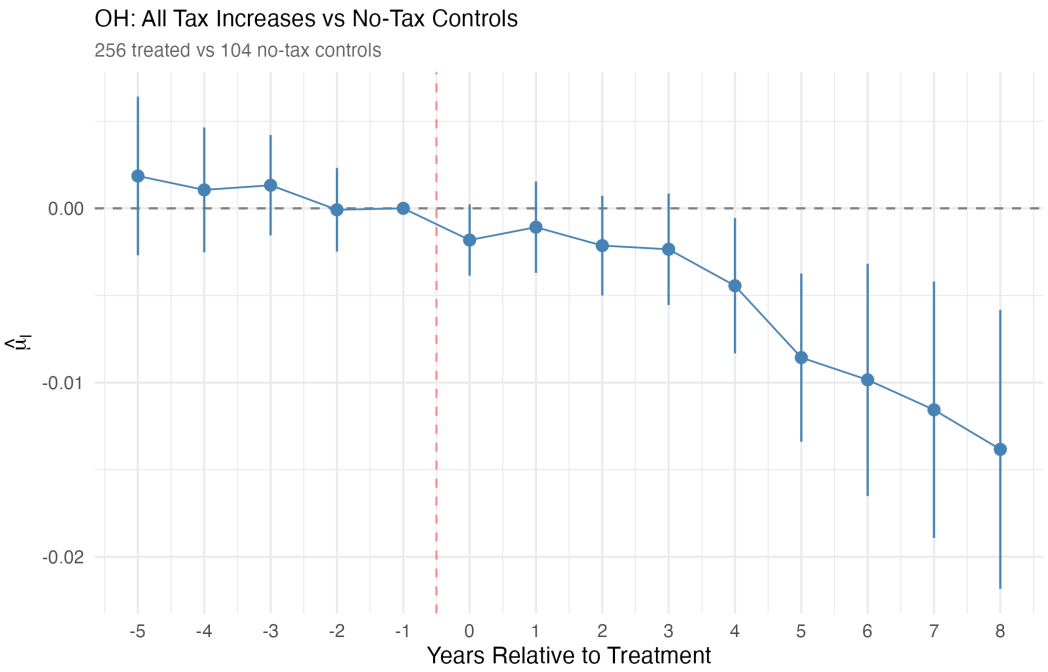


Figure 7: Q2 col (2): OH Rate Increases vs No-Tax Controls. 252 treated, 104 no-tax. Clean pre-trends; gradual decline through t+8.

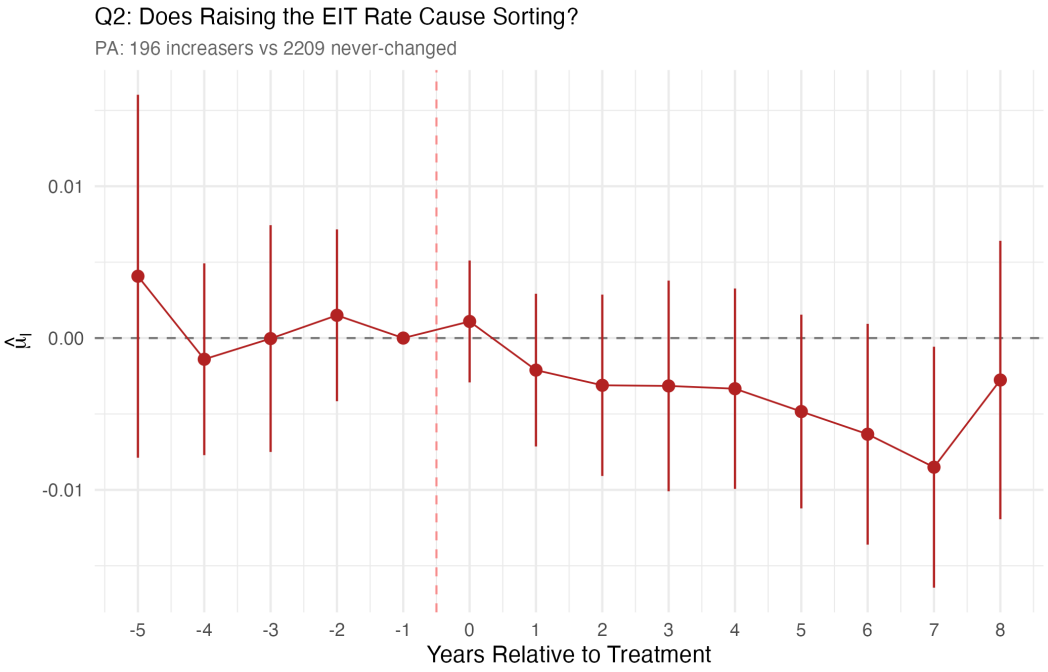


Figure 8: Q2 col (3): PA Rate Increases vs Never-Changed. 196 treated, 2,209 controls. Effect builds gradually after treatment.

5.3 Q3: Asymmetry — Do Decreases Reverse Sorting?

Table 4: Q3: Asymmetry — Increases vs Decreases

	Pennsylvania			Ohio		
	(1)	(2)	(3)	(4)	(5)	(6)
	Dec vs NC	Asymmetry	Split Dose	Dec vs NT	Asymmetry	Split Dose
Sun-Abraham ATT	0.0048			−0.0042		
	(0.0057)			(0.0064)		
increase × post		−0.0074*			−0.0055**	
		(0.0037)			(0.0020)	
decrease × post		−0.0002			−0.0080	
		(0.0060)			(0.0126)	
R^+ (cum. increase)			−0.0177*			−0.1091
			(0.0069)			(0.3788)
R^- (cum. decrease)			0.0012			−1.155
			(0.0040)			(1.353)
N	24,522	24,526	24,526	—	5,081	7,718
Fixed Effects	Municipality, Year					
Clusters	Municipality					

Notes: Columns (1), (4): Sun-Abraham ATT for decreasers vs never-changed (PA) or no-tax (OH). Columns (2), (5): joint asymmetry model with both increase and decrease indicators. PA: increases significant (-0.007^*), decreases null. OH: increases significant (-0.006^{**}), decreases null. Columns (3), (6): split cumulative dose on full sample. PA: $\beta^+ = -0.018^*$ per pp, $\beta^- \approx 0$ (asymmetric). OH split dose is imprecise due to the 0–3 rate scale. $^{\cdot}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

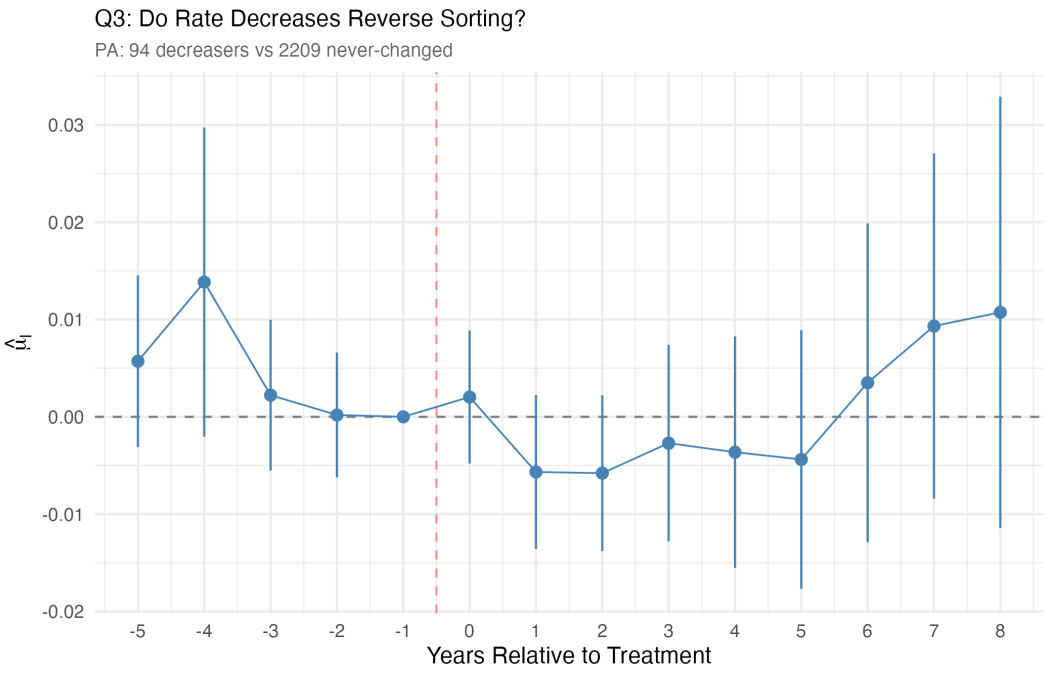


Figure 9: Q3 col (1): PA Rate Decreases vs Never-Changed. 94 treated, 2,209 controls. No evidence that rate decreases reverse sorting.

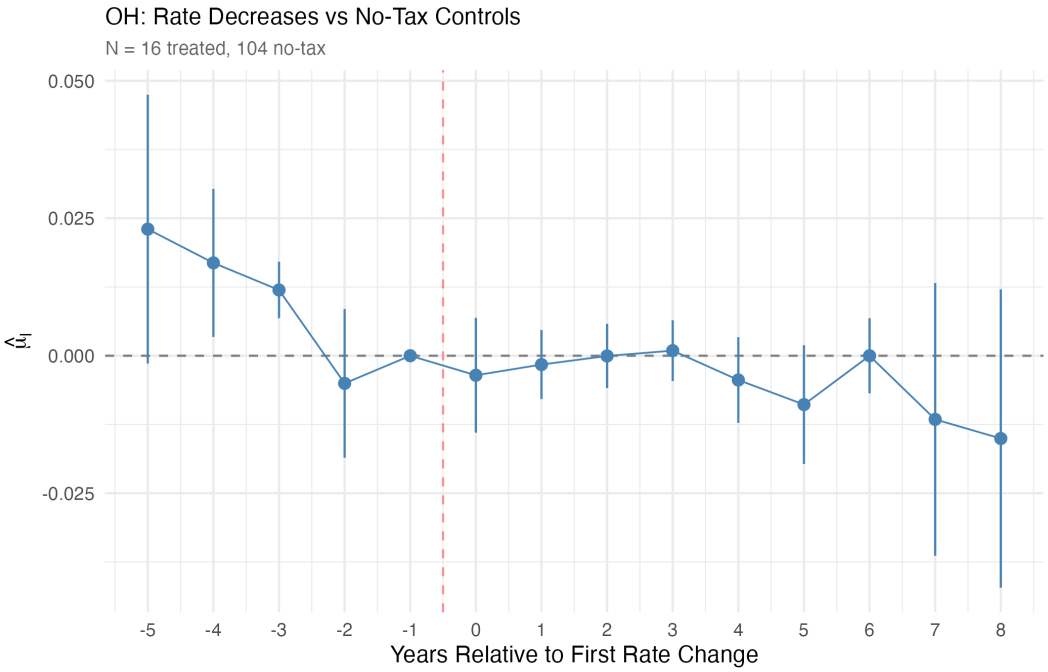


Figure 10: Q3 col (4): OH Rate Decreases vs No-Tax Controls. 36 treated, 104 controls. Wide confidence intervals; no clear pattern.

5.4 Companion Outcomes: Decomposition

Table 5: Companion Outcomes: Decomposing the share_live_in Effect

	(1)	(2)	(3)	(4)
	share_live_in	log(Resident)	log(Nonresident)	log(Total Jobs)
PA Dose-Response (treated municipalities only)				
rate × post	−0.0099·	−0.1530*	−0.1012·	−0.1172*
	(0.0052)	(0.0720)	(0.0548)	(0.0549)
post	0.0022	0.1226·	0.1113·	0.1173*
	(0.0064)	(0.0721)	(0.0596)	(0.0587)
Diagnostics				
N	2,741	2,741	2,741	2,741
R^2	0.912	0.979	0.980	0.982
Within R^2	0.011	0.015	0.008	0.011
Fixed Effects		Municipality, Year		
Clusters		Municipality		
Equality test (2) vs (3)		$p = 0.09$		

Notes: Dose-response specification estimated among PA municipalities that raised EIT rates, 2002–2019. Column (1): main outcome. Columns (2)–(4): log counts of resident workers (live and work in municipality), nonresident commuters (work in municipality, live elsewhere), and total jobs. Equality test: joint model testing whether the resident coefficient differs from the nonresident coefficient. $^{\cdot}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

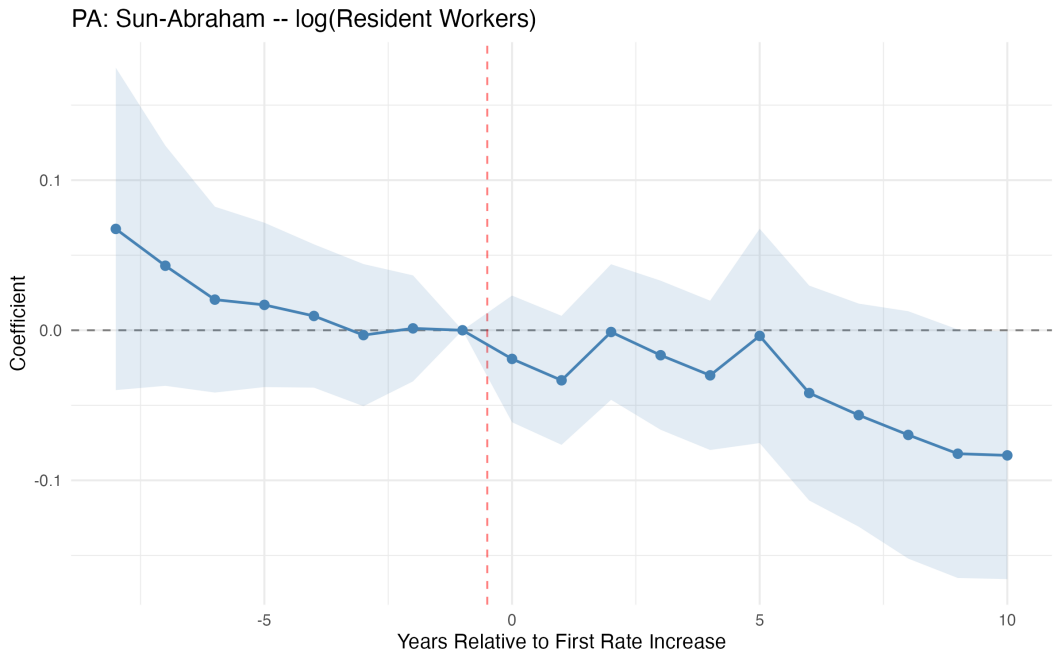


Figure 11: Pennsylvania: Sun-Abraham Event Study — log(Resident Workers).

5.5 Fiscal Impact

Table 6: Fiscal Impact of EIT Rate Changes (PA, 2002–2019)

	(1) log(EIT Rev)	(2) log(Total Tax)	(3) log(Total Rev)	(4) log(Police)	(5) log(Pop)	(6) log(Mkt Val)
resident_eit	0.6710**	0.2957**	0.0643	0.0531	−0.0094	0.0066
	(0.2106)	(0.0948)	(0.0429)	(0.0461)	(0.0137)	(0.0218)
N	42,309	42,526	42,543	22,979	42,523	9,323
R^2	0.977	0.983	0.959	0.953	0.997	0.978
Within R^2	0.027	0.006	0.0001	0.0001	0.0001	0.000
Fixed Effects	Municipality, Year					
Clusters	Municipality					

Notes: PA DCED Annual Financial Reports, 2002–2019. All outcomes in logs. Column (6) limited to 2015–2019 (market value data availability). $^{\cdot}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

5.6 Heterogeneity

6 Robustness

Table 7: Robustness: Fiscal Controls and Philadelphia Exclusion

	TWFE			Dose-Response (treated only)		
	(1)	(2)	(3)	(4)	(5)	(6)
	Baseline	+ Controls	No Philly	Baseline	+ Controls	No Philly
resident_eit	−0.0126*	−0.0104*	−0.0165**			
	(0.0052)	(0.0047)	(0.0060)			
rate × post				−0.0089**	−0.0079**	−0.0098**
				(0.0030)	(0.0028)	(0.0032)
N	27,277	14,174	27,257	3,058	2,173	3,038
Controls	No	Yes	No	No	Yes	No
Philadelphia	Yes	Yes	No	Yes	Yes	No

Notes: Controls include log police expenditure, log streets expenditure, log total debt, and log state intergovernmental aid from PA DCED Annual Financial Reports. Philadelphia excluded in columns (3) and (6) due to its structurally different wage tax system. $^{\cdot}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

6.1 Heterogeneity-Robust Continuous Treatment Estimator

As an additional robustness check, we implement the de Chaisemartin and D'Haultfoeuille (2024) estimator, which accommodates both continuous treatments and non-absorbing treatment paths under staggered timing. Unlike the Sun and Abraham (2021) estimator, which requires binary absorbing treatment, the DCDH estimator uses the actual tax rate as the treatment variable, allowing rate increases and decreases of varying magnitudes. This directly addresses the concern that our TWFE and Sun-Abraham specifications may suffer from heterogeneous treatment timing contamination when the treatment is continuous.

We estimate the DCDH specification separately for rate increases and decreases, each against never-changed controls. This mirrors the structure of our main Sun-Abraham results (Tables **??**–**??**) while handling the continuous rate as treatment.

Table 8: Robustness: de Chaisemartin–D'Haultfoeuille (2024) Estimator

	Pennsylvania		Ohio	
	(1)	(2)	(3)	(4)
	Increases	Decreases	Increases	Decreases
Average Total Effect	−0.0078	−0.0077	0.0446	2.025
	(0.0069)	(0.0090)	(0.3132)	(1.398)
Effect at $\ell = 8$	-0.0111^{***}	—	−0.0023	—
	(0.0041)		(0.0030)	
Joint effects p	0.396	0.014	0.500	0.664
Joint placebo p	0.904	0.408	0.147	0.834
Switchers	120	60	174	22
Treatment	Continuous (EIT rate / municipal rate)			
Baseline trend	Linear polynomial in period-one treatment			

Notes: De Chaisemartin and D'Haultfoeuille (2024) interaction-weighted estimator with `continuous(1)`. Treatment is the municipal income tax rate (EIT in PA, municipal rate in OH). Columns (1)–(2): PA sample restricted to increasers (decreasers) plus never-changed municipalities. Columns (3)–(4): OH, same structure. Average Total Effect is the cumulative per-unit treatment effect averaged over ∼4.3 post-treatment periods. Joint placebo p tests the null that all pre-treatment coefficients are zero. Analytical standard errors clustered at the municipality level. OH continuous estimates are imprecise due to the 0–3 rate scale. $^{\cdot}p < 0.10$, $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Table 8 reports the results. For Pennsylvania rate increases (column 1), the DCDH average total effect is −0.008 (SE 0.007), comparable to the Sun-Abraham ATT of −0.005 (SE 0.003) from Table **??**. Pre-trends are clean (joint placebo $p = 0.90$), and the effect at

$\ell = 8$ reaches -0.011 ($p < 0.001$). The PA decrease estimate (column 2) shows a negative average effect (-0.008), reinforcing the asymmetry finding: rate decreases do not reverse sorting, consistent with sticky residential location decisions.

Ohio continuous estimates (columns 3–4) are uninformative due to the 0–3 rate scale, which inflates standard errors. The Ohio results rely on the binary Sun-Abraham estimator with no-tax controls (Table **??**), where identification is sharper.

The consistency between the Sun-Abraham and DCDH estimates for Pennsylvania confirms that the main results are not driven by heterogeneous treatment timing contamination or the restriction to binary treatment.

7 Discussion

8 Conclusion

Acknowledgments

References

[1] Tiebout, C. M. (1956). A Pure Theory of Local Expenditures. *Journal of Political Economy*, 64(5), 416–424.

[2] Sun, L. & Abraham, S. (2021). Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects. *Journal of Econometrics*, 225(2), 175–199.

[3] Callaway, B. & Sant'Anna, P. H. C. (2021). Difference-in-Differences with Multiple Time Periods. *Journal of Econometrics*, 225(2), 200–230.

A Pre-Treatment Balance

Table 9: Pre-Treatment Balance: Treated vs. Never-Treated PA Municipalities

Variable	Treated Mean	Control Mean	Norm. Diff.	N (T / C)
Population	14,174	3,711	0.139	1,265 / 39,688
Total Revenue ($)	25.7M	2.9M	0.077	1,267 / 39,692
EIT Revenue ($)	4.4M	438K	0.062	
RE Tax Revenue ($)	2.9M	538K	0.129	
Police Expenditure ($)	3.1M	424K	0.110	
Streets Expenditure ($)	1.5M	496K	0.121	
EIT Rate (%)	0.50	0.50	0.055	

Notes: Treated = municipalities that raised EIT during the sample period (pre-treatment years only). Control = municipalities that never raised EIT. All normalized differences < 0.15, within standard balance thresholds. Source: PA DCED Annual Financial Reports.

B Pre-Trend Tests on Fiscal Observables

We test the parallel trends assumption on six fiscal observables from the PA DCED Annual Financial Reports using the Sun and Abraham (2021) estimator. For each outcome, we estimate Equation 1 with municipality and year fixed effects, clustering standard errors at the municipality level. Table 10 reports the maximum absolute t-statistic among the pre-treatment coefficients at relative times $\ell \in \{-5, -4, -3, -2\}$.

Pre-trends are flat for five of six fiscal observables: total revenue (max $|t| = 1.60$), EIT revenue (1.33), streets expenditure (0.80), population (1.09), and total debt (1.36). None of the pre-treatment coefficients for these outcomes are individually significant at the 5% level.

Police expenditure is the exception. The pre-treatment coefficients at $\ell = -5$ and $\ell = -4$ are negative and significant ($t = -2.85$ and $t = -2.42$, respectively), indicating that treated municipalities had relatively lower police spending 4–5 years before the rate increase.

This pattern suggests that some EIT increases may have been associated with anticipated public-safety expansions financed by the new revenue. Importantly, the main results on `share_live_in` are robust to controlling for log police expenditure (Table 7), confirming that this pre-trend does not confound the estimated sorting response.

Table 10: Pre-Trend Tests: Sun-Abraham on Fiscal Observables (PA, 2002–2019)

	Pre-Treatment Coefficients					
Outcome	$\ell = -5$	$\ell = -4$	$\ell = -3$	$\ell = -2$	Max $\lvert t \rvert$	N
log(Total Revenue)	−0.076	−0.051	−0.012	−0.052	1.60	43,170
	(0.049)	(0.056)	(0.038)	(0.032)		
log(EIT Revenue)	−0.025	−0.033	0.027	0.049	1.33	42,278
	(0.051)	(0.052)	(0.032)	(0.037)		
log(Police Exp)	-0.085^{***}	-0.068^{**}	0.011	0.037	2.85	23,464
	(0.030)	(0.028)	(0.052)	(0.039)		
log(Streets Exp)	−0.001	0.003	−0.043	−0.029	0.80	43,105
	(0.059)	(0.075)	(0.053)	(0.059)		
log(Population)	0.011	0.007	0.002	0.003	1.09	43,150
	(0.012)	(0.008)	(0.006)	(0.003)		
log(Total Debt)	−0.029	−0.051	−0.179	−0.103	1.36	24,863
	(0.176)	(0.214)	(0.132)	(0.102)		

Notes: Each row reports Sun and Abraham (2021) estimates of the pre-treatment coefficients $\hat{\mu}_\ell$ for $\ell \in \{-5, -4, -3, -2\}$, with $\ell = -1$ as the reference period. Standard errors (in parentheses) are clustered at the municipality level. Max $|t|$ is the largest absolute t-statistic among the four pre-treatment coefficients. Pre-trends are flat for all outcomes except police expenditure, where treated municipalities had significantly lower spending at $\ell = -5$ and $\ell = -4$. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Source: PA DCED Annual Financial Reports.

C Additional Figures